

SPECTRAL SEQUENCES FOR SHEAF COHOMOLOGY

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ABSTRACT. These notes were prepared for a seminar on *coherent sheaves and their parameter spaces* at the University of Pennsylvania in Fall 2025. We construct the hypercohomology spectral sequence and the Grothendieck spectral sequence, and outline important consequences.

0.1. Preliminaries. We begin by recalling the basics of spectral sequences.

Definition 0.1. Given an abelian category $\mathcal{A}$, a cohomological spectral sequence is a sequence $\{E_r^{pq}, d_r^{pq}\}$ of objects E_r^{pq} and differentials $d_r^{pq} : E_r^{pq} \rightarrow E_r^{p+r, q-r+1}$ satisfying

- (a) $d_r \circ d_r = 0$,
- (b) $E_{r+1}^{pq} \cong \ker d_r^{pq} / \operatorname{im} d_r^{p-r, q+r-1}$.

A spectral sequence stabilizes if $E_r^{pq} = E_{r+1}^{pq} = \dots$ for all sufficiently large r ; this stable value is denoted E_∞^{pq} . A spectral sequence is *weakly convergent* if E_∞^{pq} is the associated graded of a filtered object; that is, if there exists an object $H^\bullet$ with a decreasing filtration $F^\bullet$ such that

$$E_\infty^{pq} = F^p H^{p+q} / F^{p+1} H^{p+q} = gr^p H^{p+q}$$

This is written $E_r^{pq} \implies H^{p+q}$. A spectral sequence *converges* if the filtration is exhaustive and separated: $\cap_p F^p H^n = 0$ and $\cup_p F^p H^n = H^n$.

Example 0.2. We recall the construction of a spectral sequence on a filtered complex. Let $C^\bullet$ be a complex in $ch(\mathcal{A})$ with a decreasing filtration $F^\bullet$ satisfying $d(F^p C^n) \subset F^p C^{n+1}$. We define r -almost cycles and r -almost boundaries

$$Z_r^{p,q} := \ker(d : F^p C^{p+q} / F^{p+1} C^{p+q} \longrightarrow F^p C^{p+q+1} / F^{p+r} C^{p+q+1})$$

$$B_r^{p,q} := \operatorname{im}(d : F^{p-r+1} C^{p+q-1} / F^{p-r+2} C^{p+q-1} \longrightarrow F^p C^{p+q} / F^{p+1} C^{p+q})$$

and then we let $E_r^{pq} = Z_r^{pq} / B_r^{pq}$, with differentials d_r^{pq} given as the restriction of cochain differentials d . If this filtration is exhaustive and separated, then

Lemma 0.3. $\{E_r^{pq}, d_r^{pq}\}$ is a spectral sequence with $E_r^{pq} \implies h^{p+q}(C^\bullet)$ and

$$E_0^{pq} = gr^p C^{p+q}$$

$$E_1^{p,q} = h^{p+q}(gr^p C^\bullet)$$

0.2. Filtered Double Complexes. Let $C^{\bullet,\bullet}$ be a first quadrant double complex in $ch^2(\mathcal{A})$. Recall that the totalization $tot(C)^{\bullet}$ is the complex defined by

$$tot(C)^n = \bigoplus_{i+j=n} C^{i,j}$$

with differential $d^n|_{C^{i,j}} = d_1^{ij} + (-1)^i d_2^{ij}$. To orient ourselves, suppose that d_1 travels in the horizontal direction and d_2 travels in the vertical direction. Note that there are two natural filtrations on $tot(C)^{\bullet}$:

$$F_{(1)}^p tot(C)^n = \bigoplus_{r \geq p} C^{r, n-r}$$

$$F_{(2)}^p tot(C)^n = \bigoplus_{r \geq p} C^{n-r, r}$$

The former is a filtration in the vertical direction, and the latter in the horizontal direction. We define complexes $h_1^{\bullet,j}(C)$ and $h_2^{i,\bullet}(C)$ by

$$h_1^{\bullet,j}(C) = \ker d_2^{ij} / \operatorname{im} d_2^{i,j-1}$$

$$h_2^{i,\bullet}(C) = \ker d_1^{ij} / \operatorname{im} d_1^{i-1,j}$$

Caution 0.4. The former is the complex arising from taking the j -th vertical cohomology of the columns and hence travels in the horizontal direction. The latter is the i -th horizontal cohomology of the rows and hence travels in the vertical direction.

Proposition 0.5. *By 0.3, there are spectral sequences $^{(1)}E_r^{pq}$ and $^{(2)}E_r^{pq}$ with*

$$\begin{array}{ll} ^{(1)}E_0^{pq} = C^{pq} & ^{(2)}E_0^{pq} = C^{qp} \\ ^{(1)}E_1^{pq} = h_1^{pq}(C) & ^{(2)}E_1^{pq} = h_2^{qp}(C) \\ ^{(1)}E_2^{pq} = h^p(h_1^{\bullet,q}(C)) & ^{(2)}E_2^{pq} = h^p(h_2^{q,\bullet}(C)) \end{array}$$

both of which converge to $h^{p+q}(tot(C)^{\bullet})$.

Remark 0.6. In what follows, we primarily use $^{(2)}E_2^{pq}$.

0.3. Hypercohomology. Recall that if $F : \mathcal{A} \rightarrow \mathcal{B}$ is a left exact functor between abelian categories, where $\mathcal{A}$ has enough injectives, and $C^{\bullet}$ is a complex in $ch(\mathcal{A})$, the n -th right hyperderived functors of F are defined by

$$\mathbf{R}^n F(C^{\bullet}) = h^n(F(tot(I)^{\bullet}))$$

where $C^{\bullet} \rightarrow I^{\bullet,\bullet}$ is an injective resolution.

Proposition 0.5 extends to a spectral sequence computing hyperderived functors.

Proposition 0.7. *There is a spectral sequence E_r^{pq} such that*

$$E_2^{pq} = R^p F(h^q(C^{\bullet})) \implies \mathbf{R}^{p+q} F(C^{\bullet})$$

which allows us to compute hyperderived functors in terms of derived functors.

Sketch of Proof. Take I to be a Cartan-Eilenberg resolution of C .¹ The spectral sequence ${}^{(2)}E_2^{pq}$ in 0.5 for the double complex $F(I)^{\bullet,\bullet}$ yields the desired spectral sequence. The main idea is that $h^q(C^\bullet) \rightarrow h_2^{q,\bullet}(I)$ is an injective resolution, so

$$R^p F(h^q(C^\bullet)) = h^p(F(h_2^{q,\bullet}(I)))$$

by definition. Since I is Cartan-Eilenberg, there is a natural isomorphism

$$h^p(F(h_2^{q,\bullet}(I))) = h^p(h_2^{q,\bullet}(FI))$$

and we observe that the right-hand side is precisely ${}^{(2)}E_2^{pq}$. Then 0.5 implies

$${}^{(2)}E_2^{pq} \implies h^{p+q}(\text{tot}(FI)^\bullet)$$

noting that by the additivity of F the right-hand side is $\mathbf{R}^{p+q}F(C)$. $\square$

0.4. Grothendieck Spectral Sequence. An important consequence of this result is a spectral sequence for the composition of derived functors. Consider left exact additive functors $F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{C}$ where $\mathcal{A}$ and $\mathcal{B}$ have enough injectives, and such that F sends injective objects to G -acyclic objects $(\star)$.

Remark 0.8. In the following derivation, we employ the machinery of total derived functors and derived categories. Recall that the total derived functor

$$RF : D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$$

is a functor between the (bounded below) derived category of $\mathcal{A}$ and that of $\mathcal{B}$. There is a natural isomorphism of functors $h^i(RF) \cong R^i F$ and $R(G \circ F) \cong RG \circ RF$.

Let K be an object of $\mathcal{A}$, and consider the total derived functor $RF(K)$, which has a representative in $ch(\mathcal{B})$. Applying the hypercohomology spectral sequence,

$$\begin{array}{ccc} E_2^{pq} = R^p G(h^q(RF(K))) & \implies & h^{p+q}(RG(RF(K))) \\ \parallel & & \parallel \\ R^p G(R^q F(K)) & & R^{p+q}(G \circ F)(K) \end{array}$$

We define the above to be the *Grothendieck spectral sequence* for F and G .

0.5. Leray Spectral Sequence. Let $f : X \rightarrow Y$ be a continuous map of topological spaces and consider the induced functors

$$Shv(X) \xrightarrow{f_*} Shv(Y) \xrightarrow{\Gamma} Ab$$

Note that $\Gamma(X, \mathcal{F}) = \Gamma(Y, f_* \mathcal{F})$. As shown previously, the category of sheaves of abelian groups on X contains enough injective objects. The following lemma shows that f_* preserves injective objects in view of the adjunction

$$f^{-1} \dashv f_*$$

¹A Cartan-Eilenberg resolution for an object exists provided that $\mathcal{A}$ has enough injectives.

with $f^{-1} : Shv(Y) \rightarrow Shv(X)$ exact. This last fact follows since a sequence of sheaves is exact if and only if it is exact on stalks and $f^{-1}(\mathcal{G})_x \cong \mathcal{G}_{f(x)}$.

Lemma 0.9. *If $L : \mathcal{B} \rightarrow \mathcal{A}$ is exact with right adjoint R , then R preserves injectives.*

Proof. Suppose that A is an injective object of $\mathcal{A}$. Given a monomorphism $B \rightarrow B'$ and a morphism $B \rightarrow R(A)$, apply the exact functor L to obtain a monomorphism $L(B) \rightarrow L(B')$ and a morphism $L(B) \rightarrow LR(A)$. The latter maps to A via the counit of the adjunction $LR \rightarrow \text{id}_{\mathcal{A}}$. Since A is injective, there exists an extension $L(B') \rightarrow A$. Applying R yields $RL(B') \rightarrow R(A)$. This gives a compatible map $B' \rightarrow R(A)$ via the unit, which completes the proof. $\square$

Thus the image of f_* is Γ -acyclic. Consequently, hypothesis $(\star)$ is satisfied. Therefore the Grothendieck spectral sequence for the composition yields

$$E_2^{pq} = H^p(Y, R^q f_* \mathcal{F}) \implies H^{p+q}(X, \mathcal{F})$$

which is the *Leray spectral sequence*.

Example 0.10. (Higher Direct Images) Consider the composition

$$Shv(X) \xrightarrow{i} PShv(X) \xrightarrow{f_*} PShv(Y) \xrightarrow{sh} Shv(Y)$$

where $i : Shv(X) \rightarrow PShv(X)$ is the forgetful functor and $sh : PShv(Y) \rightarrow Shv(Y)$ denotes sheafification. Since the forgetful functor is right adjoint to sheafification, it is left exact. Moreover, at the level of presheaves, the functors f_* and sh are *exact*. Thus for a sheaf $\mathcal{F}$ on X ,

$$R^q f_* \mathcal{F} = R^q (sh \circ f_* \circ i)(\mathcal{F}) = (sh \circ f_*)(R^q i_* \mathcal{F}) \quad (1)$$

For an open $U \subset X$, consider the composition $Shv(X) \rightarrow PShv(X) \rightarrow Ab$ of $\Gamma_U \circ i$. Then

$$H^q(U, \mathcal{F}) = R^q(\Gamma_U \circ i)(\mathcal{F}) = \Gamma(U, R^q i_* \mathcal{F}) \quad (2)$$

again by the exactness of Γ_U for presheaves. We conclude that $R^q f_* \mathcal{F}$ is the sheafification of the presheaf

$$U \mapsto H^q(f^{-1}U, \mathcal{F})$$

by (1) and (2).

0.6. Čech-to-cohomology spectral sequence. Let $\mathfrak{U} = \{U_i\}_{i \in I}$ be a covering of a topological space X . We define a functor $\check{H}^0(\mathfrak{U}, -) : PShv(X) \rightarrow Ab$ by

$$\mathcal{F} \mapsto \text{eq} \left(\prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j) \right)$$

One can verify that $\check{H}^0(\mathfrak{U}, -)$ is additive and left exact. We define

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) := R^p \check{H}^0(\mathfrak{U}, \mathcal{F})$$

to be the p -th Čech cohomology of $\mathfrak{U}$ with values in the presheaf $\mathcal{F}$. Note that $\check{H}^0(\mathfrak{U}, \mathcal{F}) = \mathcal{F}(X)$, and therefore the composition

$$Shv(X) \xrightarrow{i} PShv(X) \xrightarrow{\check{H}^0(\mathfrak{U}, -)} Ab$$

sends injective objects to $\check{H}^0(\mathfrak{U}, -)$ -acyclic objects. Consequently, hypothesis $(\star)$ is satisfied, and there is a Grothendieck spectral sequence

$$\check{H}^p(\mathfrak{U}, R^q i_* \mathcal{F}) \implies H^{p+q}(X, \mathcal{F})$$

abutting to the standard cohomology of X valued in $\mathcal{F}$.

Exercise 0.11. If $\mathfrak{U} = \{U, V\}$, show that this spectral sequence degenerates at E_2 and yields the Mayer-Vietoris sequence:

$$0 \rightarrow H^0(X, \mathcal{F}) \rightarrow H^0(U, \mathcal{F}) \oplus H^0(V, \mathcal{F}) \rightarrow H^0(U \cap V, \mathcal{F}) \rightarrow H^1(X, \mathcal{F}) \rightarrow \dots$$

REFERENCES

- [Tam94] G. Tamme, *Introduction to Étale Cohomology*, Universitext, Springer-Verlag, Berlin, 1994, Translated from the German by Manfred Kolster.
- [Tor] G. D. Torres, *The Grothendieck Spectral Sequence*, Lecture notes, University of Texas at Austin, available at <https://web.ma.utexas.edu/users/gdavor/notes/gss.pdf>.
- [Wei94] C. A. Weibel, *An Introduction to Homological Algebra*, Cambridge Studies in Advanced Mathematics, vol. 38, Cambridge University Press, Cambridge, 1994.